

Solution fiche tp 07

SOLUTION EXO 01 :

```
#include <stdio.h>

int main() {
    int i,j,n;
    printf("Entrer l'ordre de la matrice : ");
    scanf("%d", &n);
    int A[n][n];
    int somme = 0;
    for( i = 0; i < n; i++) {
        for( j = 0; j < n; j++) {
            printf("A[%d][%d] = ", i, j);
            scanf("%d", &A[i][j]);
        }
    }
    for( i = 0; i < n; i++) {
        for( j = 0; j < n; j++) {
            printf(" %d ", A[i][j]);
        }printf("\n ");
    }
    for( i = 0; i < n; i++) {
        somme += A[i][i];
    }
    printf("Somme de la diagonale = %d\n", somme);
    return 0;
}
```

```
C:\Users\PC\Documents\test.exe
Entrer l'ordre de la matrice : 3
A[0][0] = 1
A[0][1] = 0
A[0][2] = 0
A[1][0] = 0
A[1][1] = 1
A[1][2] = 0
A[2][0] = 0
A[2][1] = 0
A[2][2] = 1
1 0 0
0 1 0
0 0 1
Somme de la diagonale = 3
-----
Process exited after 10.09 seconds with
Press any key to continue . . .
```

SOLUTION EXO 02

```
#include <stdio.h>

int main() {
    int n,i,j;
    printf("Entrer l'ordre de la matrice : ");
    scanf("%d", &n);
    int A[n][n], L[n], C[n];
    for( i = 0; i < n; i++) {
        L[i] = 0;
        C[i] = 0;
    }
    for( i = 0; i < n; i++) {
        for( j = 0; j < n; j++) {
            printf("A[%d][%d] = ", i, j);
            scanf("%d", &A[i][j]);
            L[i] += A[i][j];
        }
        C[j] += A[i][j];
    }
    printf("\nSomme des lignes : ");
    for( i = 0; i < n; i++) printf("%d ", L[i]);
    printf("\nSomme des colonnes : ");
    for( j = 0; j < n; j++) printf("%d ", C[j]);
    return 0;
}
```

```
C:\Users\PC\Documents\exo02tp7.exe
Entrer l'ordre de la matrice : 4
A[0][0] = 1
A[0][1] = 2
A[0][2] = 3
A[0][3] = 4
A[1][0] = 5
A[1][1] = 6
A[1][2] = 6
A[1][3] = 5
A[2][0] = 4
A[2][1] = 3
A[2][2] = 2
A[2][3] = 1
A[3][0] = 1
A[3][1] = 2
A[3][2] = 3
A[3][3] = 4
Somme des lignes : 10 22 10 10
Somme des colonnes : 11 13 14 14
-----
Process exited after 16.36 seconds with
Press any key to continue . . .
```

SOLUTION EXO 03

```
#include <stdio.h>
int main() {
    int n, m, i, j, k;
    printf("Entrez n et m : ");
    scanf("%d %d", &n, &m);
    int A[n][m], B[m][n], C[n][n];
    printf("Entrez A (%d x %d) :\n", n, m);
    for ( i = 0; i < n; i++)
        for ( j = 0; j < m; j++)
            scanf("%d", &A[i][j]);
    printf("Entrez B (%d x %d) :\n", m, n);
    for ( i = 0; i < m; i++)
        for ( j = 0; j < n; j++)
            scanf("%d", &B[i][j]);
    for ( i = 0; i < n; i++)
        for ( j = 0; j < n; j++)
            C[i][j] = 0;
    for ( i = 0; i < n; i++)
        for ( j = 0; j < n; j++)
            for ( k = 0; k < m; k++)
                C[i][j] += A[i][k] * B[k][j];
    printf("Matrice C (%d x %d) :\n", n, n);
    for ( i = 0; i < n; i++) {
        for ( j = 0; j < n; j++)
            printf("%d ", C[i][j]);
        printf("\n");
    }
    return 0;
}
```

```
C:\Users\PC\Documents\3
Entrez n et m : 3 2
Entrez A (3 x 2) :
4 2
6 5
8 7
Entrez B (2 x 3) :
4 5 6
2 3 4
Matrice C (3 x 3) :
20 26 32
34 45 56
46 61 76

-----
Process exited after 23
Press any key to contin
```

SOLUTION EXO 03 avec une petite modification dans le code pour expliquer la methode de calcul

```
#include <stdio.h>
int main() {
    int n, m, i, j, k;
    printf("Entrez n et m : ");
    scanf("%d %d", &n, &m);
    int A[n][m], B[m][n], C[n][n];
    printf("Entrez A (%d x %d) :\n", n, m);
    for ( i = 0; i < n; i++)
        for ( j = 0; j < m; j++)
            scanf("%d", &A[i][j]);
    printf("Entrez B (%d x %d) :\n", m, n);
    for ( i = 0; i < m; i++)
        for ( j = 0; j < n; j++)
            scanf("%d", &B[i][j]);
    for ( i = 0; i < n; i++)
        for ( j = 0; j < n; j++)
            C[i][j] = 0;
    for ( i = 0; i < n; i++)
        for ( j = 0; j < n; j++){
            for ( k = 0; k < m; k++){
                C[i][j] += A[i][k] * B[k][j];
                printf(" C[%d][%d]= %d * %d \n", i, j, A[i][k], B[k][j] );
            }
            printf(" C[%d][%d]= %d \n", i, j, C[i][j] );
        }
    printf("Matrice C (%d x %d) :\n", n, n);
    for ( i = 0; i < n; i++) {
        for ( j = 0; j < n; j++)
            printf("%d ", C[i][j]);
        printf("\n");
    }
}
```

```
C:\Users\PC\Documer
Entrez n et m : 3 2
Entrez A (3 x 2) :
3 4
2 1
5 6
Entrez B (2 x 3) :
1 1 1
1 1 1
C[0][0]= 3 * 1
C[0][0]= 4 * 1
C[0][0]= 7
C[0][1]= 3 * 1
C[0][1]= 4 * 1
C[0][1]= 7
C[0][2]= 3 * 1
C[0][2]= 4 * 1
C[0][2]= 7
C[1][0]= 2 * 1
C[1][0]= 1 * 1
C[1][0]= 3
C[1][1]= 2 * 1
C[1][1]= 1 * 1
C[1][1]= 3
C[1][2]= 2 * 1
C[1][2]= 1 * 1
C[1][2]= 3
C[2][0]= 5 * 1
C[2][0]= 6 * 1
C[2][0]= 11
C[2][1]= 5 * 1
C[2][1]= 6 * 1
C[2][1]= 11
C[2][2]= 5 * 1
C[2][2]= 6 * 1
C[2][2]= 11
Matrice C (3 x 3) :
7 7 7
3 3 3
11 11 11

-----
Process exited after 16.58
Press any key to continue
```

SOLUTION EXO 04

```
#include <stdio.h>
int main() {
    int n, m;
    printf("Entrer n et m : ");
    scanf("%d %d", &n, &m);
    int A[n][m];
    int i, j, min, max;
    int imin, jmin, imax, jmax;
    min = max = A[0][0];
    imin = imax = 0;
    jmin = jmax = 0;
    for( i = 0; i < n; i++) {
        for( j = 0; j < m; j++) {
            printf("a[%d][%d]", i, j);
            scanf("%d", &A[i][j]);

            if(A[i][j] < min) { min = A[i][j]; imin = i; jmin = j; }
            if(A[i][j] > max) { max = A[i][j]; imax = i; jmax = j; }
        }
    }
    printf("\n Min = %d position (%d,%d)\n", min, imin, jmin);
    printf("Max = %d position (%d,%d)\n", max, imax, jmax);
    return 0;
}
```

```
C:\Users\PC\Documents\ex3tp7.exe
Entrer n et m : 3
3
a[0][0]7
a[0][1]8
a[0][2]77
a[1][0]0
a[1][1]5
a[1][2]4
a[2][0]3
a[2][1]5
a[2][2]8

Min = 0 position (1,0)
Max = 77 position (0,2)
```

SOLUTION EXO 05

```
#include <stdio.h>
int main() {
    int n, m, i, j;
    printf("Entrez n et m : ");
    scanf("%d %d", &n, &m);
    int A[n][m], T[m][n];
    printf("Entrez la matrice A :\n");
    for ( i = 0; i < n; i++)
        for ( j = 0; j < m; j++)
            scanf("%d", &A[i][j]);
    for ( i = 0; i < n; i++)
        for ( j = 0; j < m; j++)
            T[j][i] = A[i][j];
    printf("Matrice transposée :\n");
    for ( i = 0; i < m; i++) {
        for ( j = 0; j < n; j++)
            printf("%d ", T[i][j]);
        printf("\n");
    }
    return 0;
}
```

```
C:\Users\PC\Documents\3wv
Entrez n et m : 3 2
Entrez la matrice A :
4 3
7 5
3 2
Matrice transposée :
4 7 3
3 5 2

-----
Process exited after 26.1
Press any key to continue
```